

HOLMES REEF, AustraliaWMO: **942890**Lat: **16.468S**Lon: **147.873E**Elev: **6**StdP: **101.25**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.4	22.0	N/A	N/A	N/A	N/A	N/A	N/A	14.3	23.1	13.6	23.3	8.4	130	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.6	30.1	N/A	29.6	N/A	29.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	13.1	12.3	20.3	31.3	1.4	0.6	19.3	31.8	18.4	32.1	17.6	32.5	16.6	33.0		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	27.9	28.1	27.6	26.7	25.6	24.3	23.6	23.5	24.2	25.3	26.4	27.5
	DBStd	1.85	0.81	0.88	0.96	0.79	0.87	0.91	0.89	0.79	0.80	0.81	0.78	0.86
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5791	554	507	544	501	482	429	420	418	426	473	493	543
	CDD18.3	2749	296	274	286	251	224	179	162	159	176	215	243	284
	CDH23.3	23543	3473	3308	3279	2521	1733	852	432	322	687	1481	2279	3178
	CDH26.7	4301	1022	1092	874	307	42	1	0	0	1	24	182	755
	WSAvg	7.1	6.1	6.0	6.8	8.4	8.4	8.4	8.2	7.7	6.9	6.6	6.1	5.9
(15) Precipitation	PrecAvg	997	160	191	211	100	59	41	31	24	13	25	39	105
	PrecMax	1571	357	429	729	292	230	118	138	104	43	237	242	400
	PrecMin	327	3	33	11	10	10	3	3	4	1	1	4	4
	PrecStd	293	106	109	142	90	57	27	31	23	10	44	54	92
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.4	30.7	30.3	28.7	28.0	26.9	26.1	25.7	26.8	28.1	29.0	30.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	29.8	30.1	29.7	28.2	27.5	26.3	25.6	25.2	26.2	27.3	28.3	29.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	29.4	29.7	29.2	28.1	27.2	26.0	25.2	24.9	25.7	26.8	27.9	29.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	29.1	29.3	29.0	27.8	26.8	25.6	25.0	24.6	25.3	26.5	27.6	28.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	1.6	1.6	1.7	1.2	1.2	1.3	1.3	1.1	1.0	1.0	1.1	1.4
		MCDBR	2.0	2.0	2.0	1.6	1.5	1.4	1.5	1.3	1.3	1.5	1.6	1.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.392	0.392	0.400	0.388	0.376	0.378	0.354	0.359	0.377	0.390	0.392	0.390
	taud		2.600	2.610	2.569	2.587	2.581	2.565	2.618	2.588	2.515	2.496	2.531	2.573
	Ebn at Noon		952	943	914	891	866	846	881	905	922	934	946	953
	Edh at Noon		105	103	104	96	90	89	86	94	108	114	111	107
(44) All-Sky Solar Radiation	RadAvg		6.55	6.29	5.67	5.34	4.63	4.14	4.46	5.36	6.43	7.06	7.28	7.05
	RadStd		0.59	0.54	0.43	0.42	0.40	0.32	0.28	0.33	0.22	0.32	0.55	0.75

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page